PREFACE

In the realm of astrophysics and data science, the quest to unravel the mysteries of the universe often hinges on the ability to accurately interpret vast amounts of data. The Gravity Spy project, a citizen science initiative aimed at classifying glitches in data from the Laser Interferometer Gravitational-Wave Observatory (LIGO), stands at the intersection of these two fields. This project has harnessed the collective power of volunteers and sophisticated machine learning models to improve the detection of gravitational waves, which are ripples in space-time caused by cataclysmic events like black hole mergers.

This report presents a comparative analysis of three advanced machine learning models—EfficientNetB2, ResNet50, and the newly proposed Attention Fusion Classifier—applied to the classification of LIGO glitches. Each model brings its unique architecture and strengths to the table, aiming to enhance the accuracy and reliability of glitch classification. EfficientNetB2 is known for its balance between performance and computational efficiency, while ResNet50 is celebrated for its deep residual learning capabilities. The Attention Fusion Classifier, designed specifically for this project, leverages inception residual blocks and an intermediate fusion strategy to capture intricate patterns and correlations in the data.

Through detailed analysis and comparison, this report not only highlights the performance metrics of each model but also delves into their architectural nuances and practical implications for gravitational wave detection. By understanding the strengths and limitations of these models, we aim to pave the way for more accurate and robust glitch classification, ultimately contributing to the advancement of gravitational wave astronomy.

We extend our gratitude to the Gravity Spy community, whose contributions have been invaluable to this endeavor. It is our hope that the insights gained from this study will inspire further innovations in the application of machine learning to astrophysical data analysis.